Counterfactual Explanations for Anomalies in Photovoltaic Power Forecasting

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1 Introduction

Photovoltaic (PV) energy is one of the fastest growing renewable power sources and has been touted as a leading alternative to a fossil fuel-based grid system [7]. However, due to its reliance on geographical, meteorological, and physical factors, it is susceptible to severe fluctuations in power output that affect a reliable large-scale implementation [25]. As such, there has been a drive towards accurate power output prediction and therefore a growing number of machine learning models are being applied for this purpose.
Recent developments have shown that the most effective of these approaches are those that rely on deep learning [25]. However, despite the remarkable accuracy of deep forecasting, these models are largely opaque and the "black-box" approach makes interpreting their results challenging [22]. This is most notable in the case of Type 1 and 2 negative anomalies: either the model's prediction is significantly above the true value, or the true value (and accurate predicted value) is significantly below its average during a local period of time. Gaining an insight into the model in the first case would allow its designers to potentially improve the accuracy of the model for outliers, while in the latter case it would provide those managing the solar farms with valuable information as to which factors most significantly contributed to a drop in power output, and, possibly, which could be regulated for increased efficiency, as well as building trust in the model's predictions if they agree with known information. Such insights can be gained through Explainable AI (XAI), which seeks to provide a deeper understanding of the functioning and predictions of a model [9].
In particular, this paper uses current and forecast meteorological features together with physical conditions of the solar panels to train a black-box neural network in PV output prediction before applying XAI via counterfactual explanations to the model for both types of anomalies.

2 Related Work

Counterfactual explanations have been applied to time series forecasting, with Wang et al. (2023) developing ForecastCF, an algorithm generating counterfactual explanations for regressive time series data by applying gradient-based perturbations. However, their work did not address multivariate time series forecasting. CoMTE [2] appears to fill this gap, however it is not completely model agnostic, as it can only be applied to a model that outputs probabilities associated with classifications, and not one that outputs continuous values. Moreover, to the best of my knowledge, the application of counterfactual explanations to PV power forecasting remains unexplored. With regards to explanations for PV power forecasting, various XAI techniques have been employed. Rizk-Allah et al. (2024) introduced an LSTM model and utilized LIME [1] while Kuzlu et al. (2020) applied LIME and SHAP [2] to PV power forecasting models. Both of these implementations provided valuable insights to the general workings of the model, but they do not focus on anomaly explanation in particular. Indeed, studies involving anomalies in PV power output have sought to predict them [23] [13] without analyzing the predictive model itself through XAI.
In this way, this paper provides a novel extension of counterfactual explanations into PV power output prediction and of XAI into the associated anomalies that arise from this.

[1] Local Interpretable and Model-independent Explanation
[2] SHapley Additive exPlanations

3 Dataset and Features

In order to accurately train the model on a variety of physical and meteorological features, I selected a dataset with high granularity and reliable, detailed bookkeeping. The data was collected over 2020 and 2021 from the NREL[3] 75-kw solar field in Golden, Colorado [10] with a resolution of one minute. I subsequently reduced this to one hour to reduce the amount of data and allow for faster training times while maintaining a broad time-span and good precision. In addition to the on-site weather conditions present in this dataset, I obtained historical weather forecast data from the area and added Gaussian noise to account for potential misreporting and to improve the generalizability of the model [3] [11]. The normalized features over the first two days of the data set are shown in Figure 1, where "avg_dcp" is the target variable (power output).

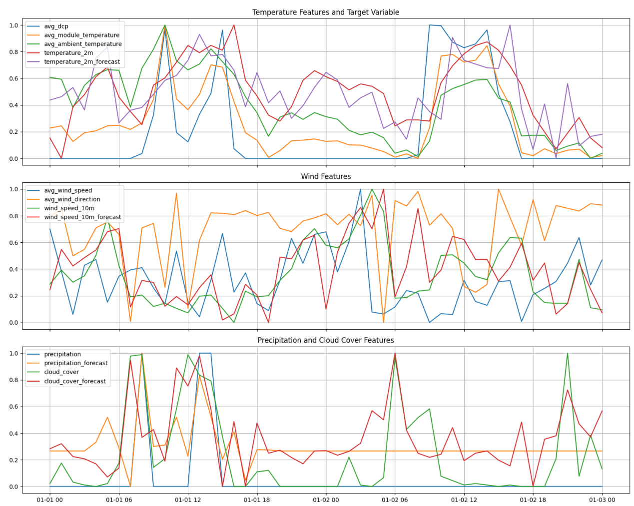

Figure 1: Normalized Features

To allow linear regression to capture cyclical dependencies, I transformed the time and the day to cosine and sine based on their position in the day and year, respectively. To ensure consistency and improve the quality of the input features, I normalized all numerical data to fall within a standard range. Finally, in order to speed up the algorithm and to group together highly correlated features, I performed PCA on the features before generating the counterfactual explanations.

4 Methodology

Evaluation Metrics: The performance of the models was evaluated using standard Root Mean Square Error (RMSE), Root Mean Absolute Error (RMAE), and the coefficient of determination (R^2). While RMSE is sensitive to large errors, highlighting deviations, RMAE provides a simpler interpretation of prediction accuracy. Finally, R^2 represents the proportion of variance accounted for by the model and is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2},$$

where values closer to 1 indicate a better fit.

Linear Regression: Linear regression was performed on the data to provide a baseline for forecast accuracy. This supplied the context to understand when the subsequent deep learning model reached a high enough accuracy to apply counterfactual explanations since the inherent simplicity of linear regression makes it easily interpretable and therefore more suitable if the other model is not significantly more accurate.

FDNet: I implemented a non linear neural network called FDNet [19], which is designed for time series forecasting. Its methodology can be broken down into the following components:

[3]National Renewable Energy Laboratory

- **Convolutional Neural Networks:** FDNet incorporates 2D convolutional layers for feature extraction, with separate kernels for temporal and variate dimensions.

- **Focal Input Sequence Decomposition:** To manage long sequence time series inputs (LSTI), this method divides the input sequence into sub-sequences based on their closeness in time to the prediction target. Closer sub-sequences are shorter and processed with deeper feature extraction layers, while farther ones are longer and handled with shallower layers. This has proven to prevent overfitting and reduce run times.

- **Decomposed Forecasting Formula:** Unlike forecasting methods that rely on global correlations, FDNet uses a decomposed formula that focuses on localized feature extraction. This enhances robustness against distribution shifts in the data, which occur in my case due to seasonal or meteorological variations, and improves the forecasting of non-stationary time series.

Counterfactual Explanations: To interpret the forecasting model's predictions, I utilized counterfactual explanations, which explore "what if" scenarios by finding minimal changes to the input $\mathbf{x}$ such that the model's output $f(\mathbf{x})$ changes to a desired target $\mathbf{y}^*$. This process is formulated as an optimization problem:

$$\min_{\mathbf{x}'} d(\mathbf{x}, \mathbf{x}') \quad \text{subject to} \quad f(\mathbf{x}') = \mathbf{y}^*,$$

where $\mathbf{x}'$ is the counterfactual instance, $d(\mathbf{x}, \mathbf{x}')$ measures the distance between the original and counterfactual instances (by some norm), and $\mathbf{y}^*$ is the target outcome. By analyzing which features were most altered in order to change a classification, one can learn which are most significant to the model's predictions, while the direction of the change can indicate in what sense they affect these forecasts. This paper implements MACE [24], a model-agnostic framework that efficiently applies these methods (partially via reinforcement learning) and can be utilized for time series forecasting [16].

5 Results

5.1 Baseline

Table 1 gives the results obtained by linear regression on the test set using the same normalized features. The results clearly show a decrease in the accuracy of the predictions over longer time spans. This agrees with the complex and non-linear relationship between environmental factors and PV power output [18].
In order to gain a basic understanding of the most significant features, the top weights for the normalized features were also recorded in Table 2. It is interesting to note that none of the predictions utilize the environmental factors to any significant degree, which is likely due to their non-linear impact on power output. The 1 hour, 24 hour, and 48 hour models place more weight on the current power output, which seems to suggest that they are just "guessing" based on the power output an hour before the prediction or at the same time 1 and 2 days before, while the mid range models seem to "guess" based on the time of day.

Time Span (Hours)	MSE	MAE	R^2 Score
1	1.408	0.7583	0.7584
6	2.0584	1.0671	0.6468
12	1.9719	1.0094	0.6617
24	1.8891	0.9024	0.6762
48	1.9559	0.9413	0.6651

Table 1: Results for Linear Regression Model Across Different Time Spans

Time Span (Hours)	Top Features	Weight
1	Current Power Output (avg_dcp)	1.4705
6	Sine of Hour (sin_hour)	0.6522
12	Cosine of Hour (cos_hour)	0.5559
24	Current Power Output (avg_dcp)	0.5410
48	Current Power Output (avg_dcp)	0.4369

Table 2: Top Feature Weights for Linear Regression Model Across Different Time Spans

5.2 FDNet

The model was trained with a backcast length of 48 hours and a forecast length of 48 hours, as well as batch sizes of 32 and epochs of 5. This remains close to what was performed in the original paper [19] and I found that it allowed for reasonable training times whilst not sacrificing accuracy. Moreover, since the model is designed to predict all the features across the forecast length, I altered the loss function to only optimize the power output prediction.

The results obtained on the test set are shown in Table 3, while Figure 2 shows an example of the predictions against the true power output. The model shows significant improvements to the baseline across all metrics, which was to be expected with its increased complexity. However, it fails to maintain its accuracy for each of the time spans, becoming less reliable as they increase, which may indicate overfitting or that the model struggles to capture long-term dependencies effectively or that critical features influencing photovoltaic power output over longer horizons are either missing or insufficiently represented in the data. This suggests the need for incorporating additional long-term environmental predictors or refining the feature engineering process to better support extended time-span forecasting.

Time Span (Hours)	MSE	MAE	R^2 Score
1	0.0192	0.1812	0.9818
6	0.1786	0.5017	0.8271
12	0.2015	0.5624	0.8094
24	0.2097	0.5837	0.7959
48	0.2624	0.6703	0.7535

Table 3: Results for FDNet Across Different Time Spans

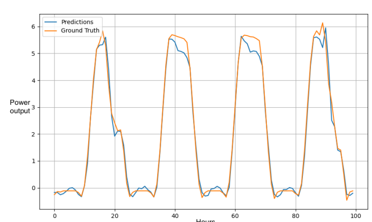

Figure 2: Predicted and True Power Output Sample

5.3 Anomalies and Explanations

After detecting anomalies in the data, counterfactual explanations were applied with the intent of identifying the smallest change in features to alter a classification of a data point from anomalous to not. While the features were originally backcast horizons over 48 hours, the explanations collated the changes for a single feature via PCA and the figures show the results for a single anomaly, since I found that they were consistent with the results for 20 other randomly sampled anomalies.

Anomalies of type 1, in which the model's prediction overestimated the true value, were identified through a deviation threshold of 0.1, which recognized approximately 10 percent of the data as anomalous. The results in Figure 2 show that for both 1 hour and 48 hour predictions, the temperature-based features experienced the largest change in value, which suggests that there would be some benefit in increasing their resolution or focusing the model's attention on them during training. Moreover, it is interesting to note that irradiance played a much more significant role in the 48 hour prediction anomalies, which could therefore also be focused on if training a model for longer time

span predictions.
Anomalies of type 2, in which the true value of power output dropped below its usual value and the model predicted this drop, were identified through their deviation from the 300 hour rolling daily mean, which helped to prevent long term seasonal changes being flagged as anomalous, and by ensuring the model's prediction was within a 0.1 threshold to the true value. The results in Figure 3 for 1 hour predictions highlight the significant impact of the temperature of the solar panel modules in identifying anomalies, with the proposed decrease agreeing with our understanding of their increased efficiency at lower temperatures [4], while the 48 hour predictions highlight the importance of humidity, irradiance, and precipitation. These factors are somewhat surprising in that it is unclear how irradiance from two days before would impact the prediction, and the increase of humidity to increase power output goes against what we know about humidity's impact on PV systems [21]. Thus, the model may be finding false correlations on the data, which may account for the model's decreased performance over longer time spans, as it is failing to acknowledge certain relationships known to people in the field.
Both sets of results underline a significant difference in the model's predictions to those of linear regression in that they reveal its reliance on environmental features which linear regression seemed to "ignore". In this sense, the model is made more trustworthy as it suggests that it is doing something more than simply extrapolating based on the time or current power output.

(a) 1 Hour Prediction

(b) 48 Hour Prediction

Figure 3: Type 1 Anomalies

(a) 1 Hour Prediction

(b) 48 Hour Prediction

Figure 4: Type 2 Anomalies

6 Conclusions and Future Work

The results presented in this work highlight the value of counterfactual explanations in helping to make sense of the predictions made by a black box model. More specifically, their insight into Type 1 anomalies exposed which features are most influential in anomalous prediction prevention and future papers might explore the impact of iteratively adapting training in response to these results. Beyond this, the application to Type 2 anomalies allowed for an insight into the factors that the model sees as contributing most greatly to a decreased power output. This granted a deeper understanding of the model's predictions and where these could be trusted or where they might be flawed, while also suggesting actionability with respect to lowering the solar power module temperatures to increase efficiency. In this way, the model acted as a form of a simulation of the solar farm, which could be interpreted to gain an insight into the factors affecting the true power output.
Future work should look into expanding the data set across more solar farms and increasing the variety of features to allow for an increasingly generalized model from which one might gain more specific explanations due to the increased features. In addition, future studies could combine SHAP with counterfactuals to generate more detailed explanations, which could be utilized in managing a simulated solar farm to determine the effectiveness of actionability based on XAI.

7 Contributions

Basic idea for the project formulated together with Haoyan Jiang (who has withdrawn from the class). The entirety of the rest of the project completed by Bodo Wirth.

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